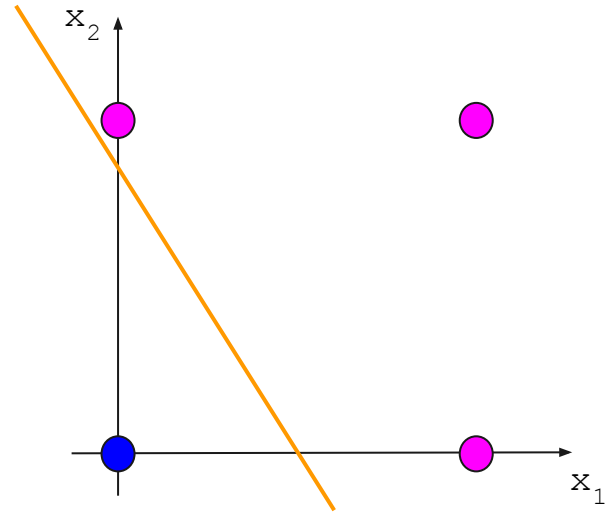
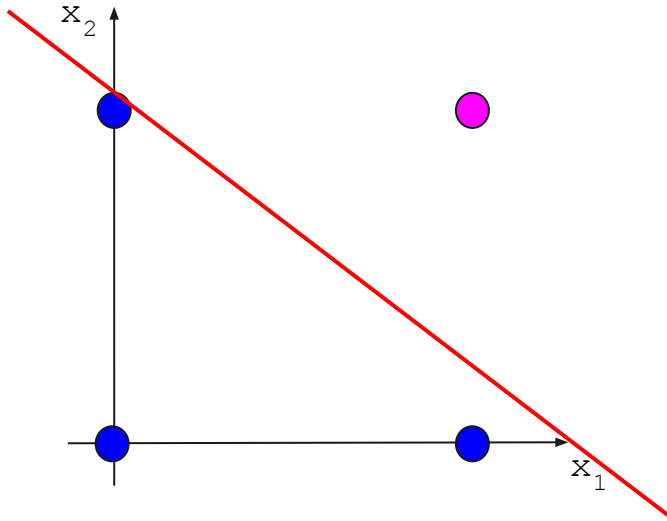




RN Multicamadas

Prof. Elder Santos

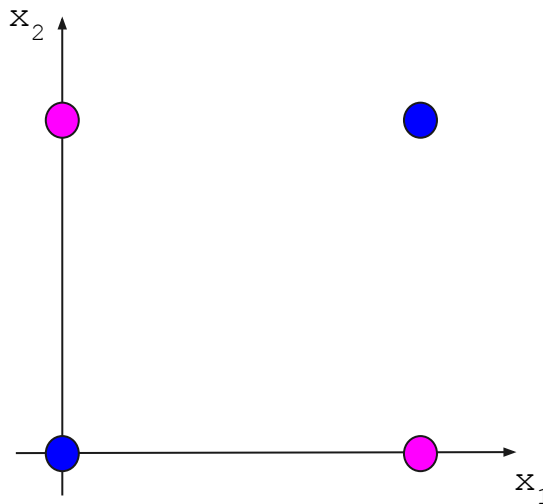
Resultado Neurônios AND e OR



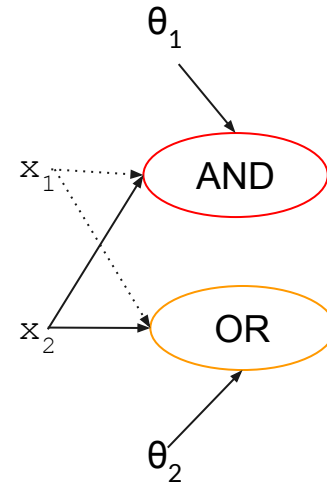
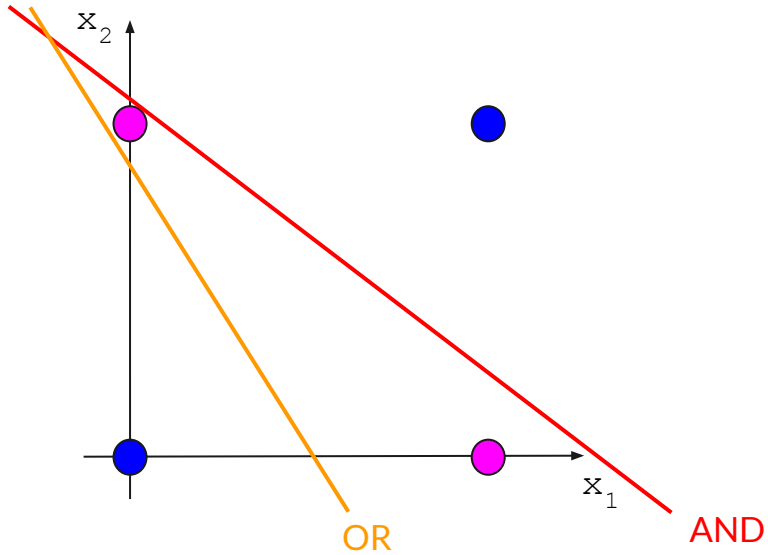
Exemplo XOR

Não é linearmente separável

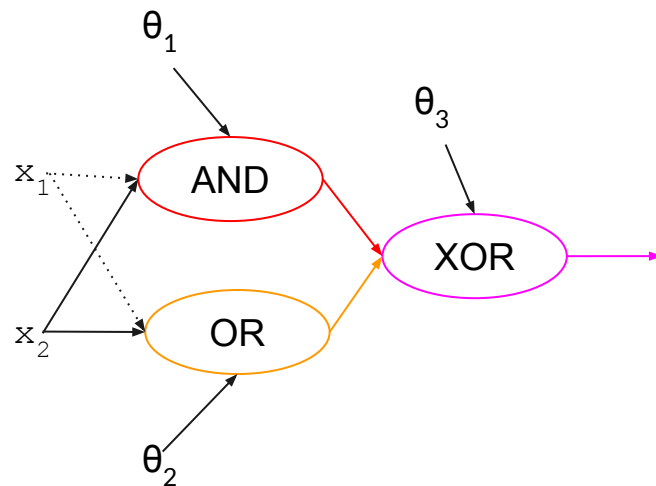
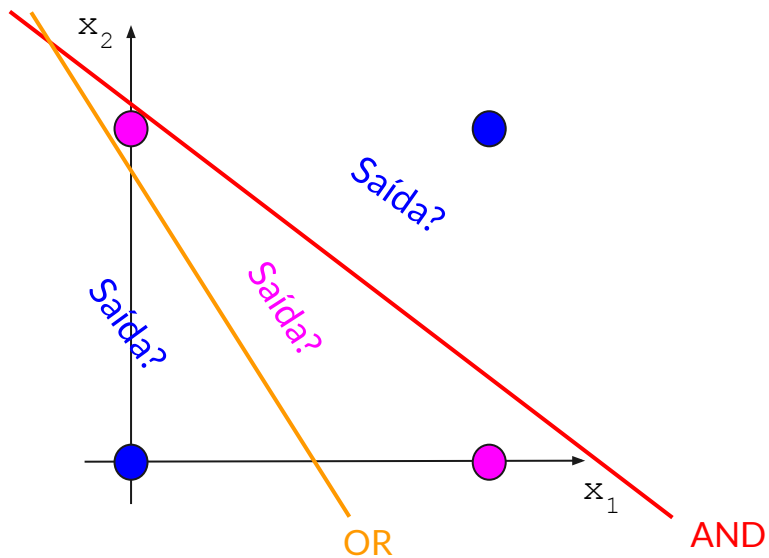
Precisamos de duas retas
para separar os grupos



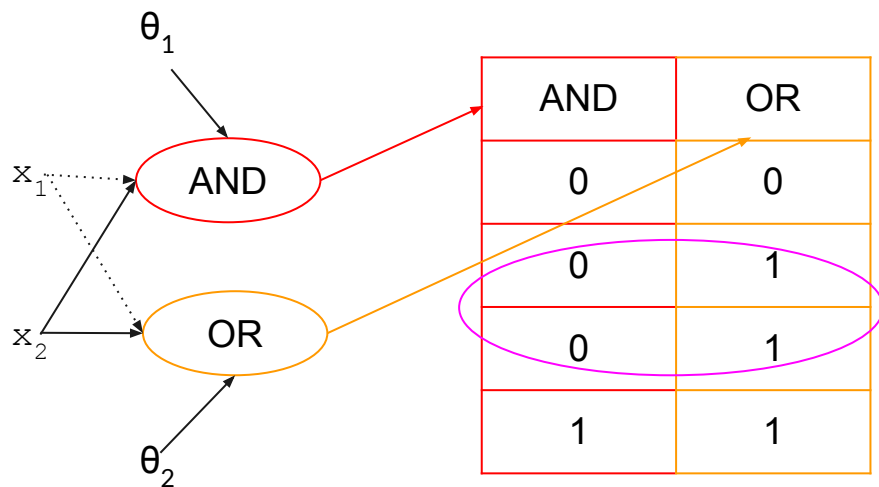
Exemplo XOR



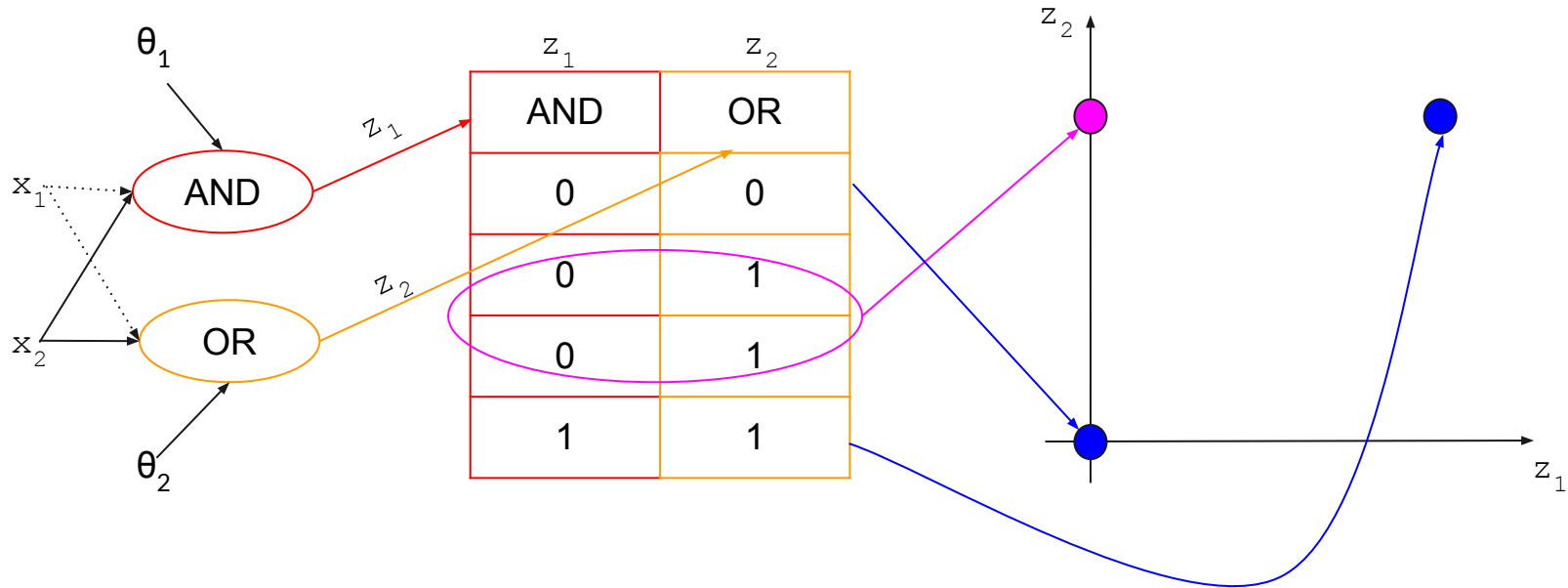
Exemplo XOR



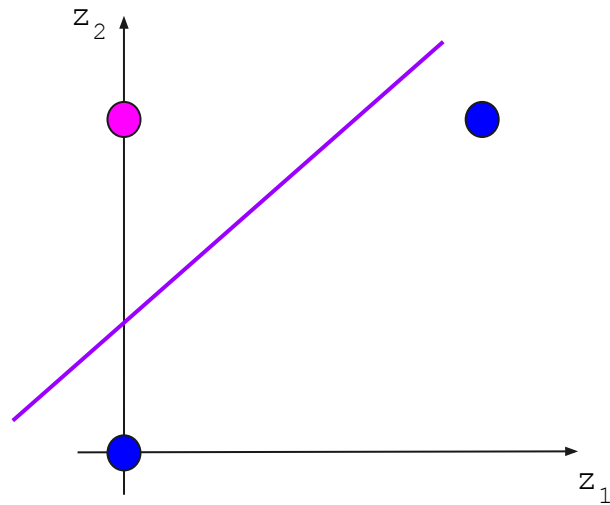
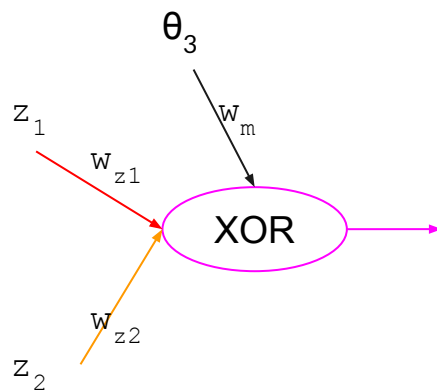
Resultado Camada Oculta



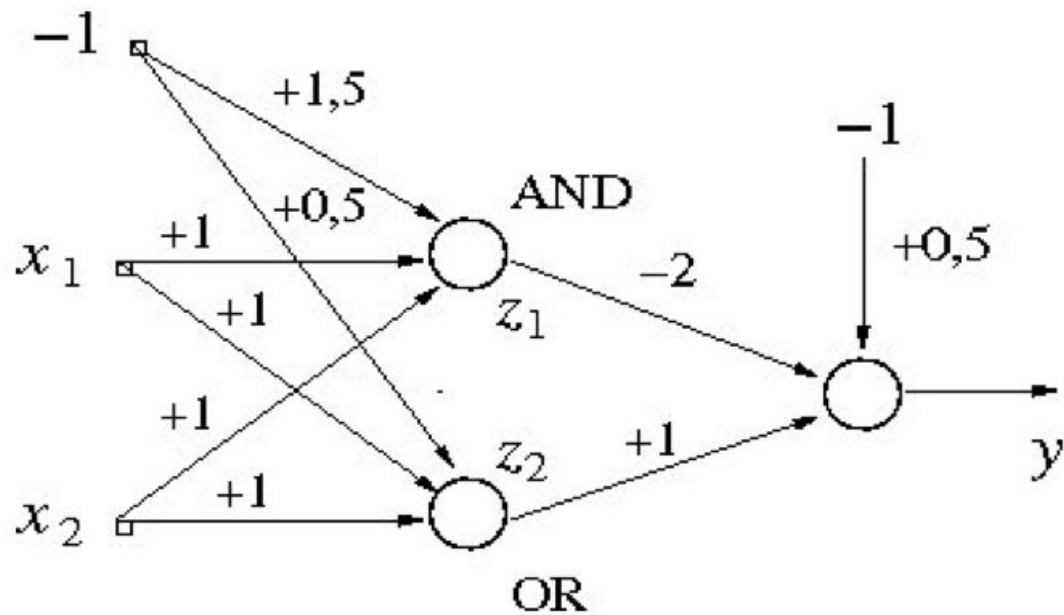
Resultado Camada Oculta



Camada de Saída



Rede para XOR



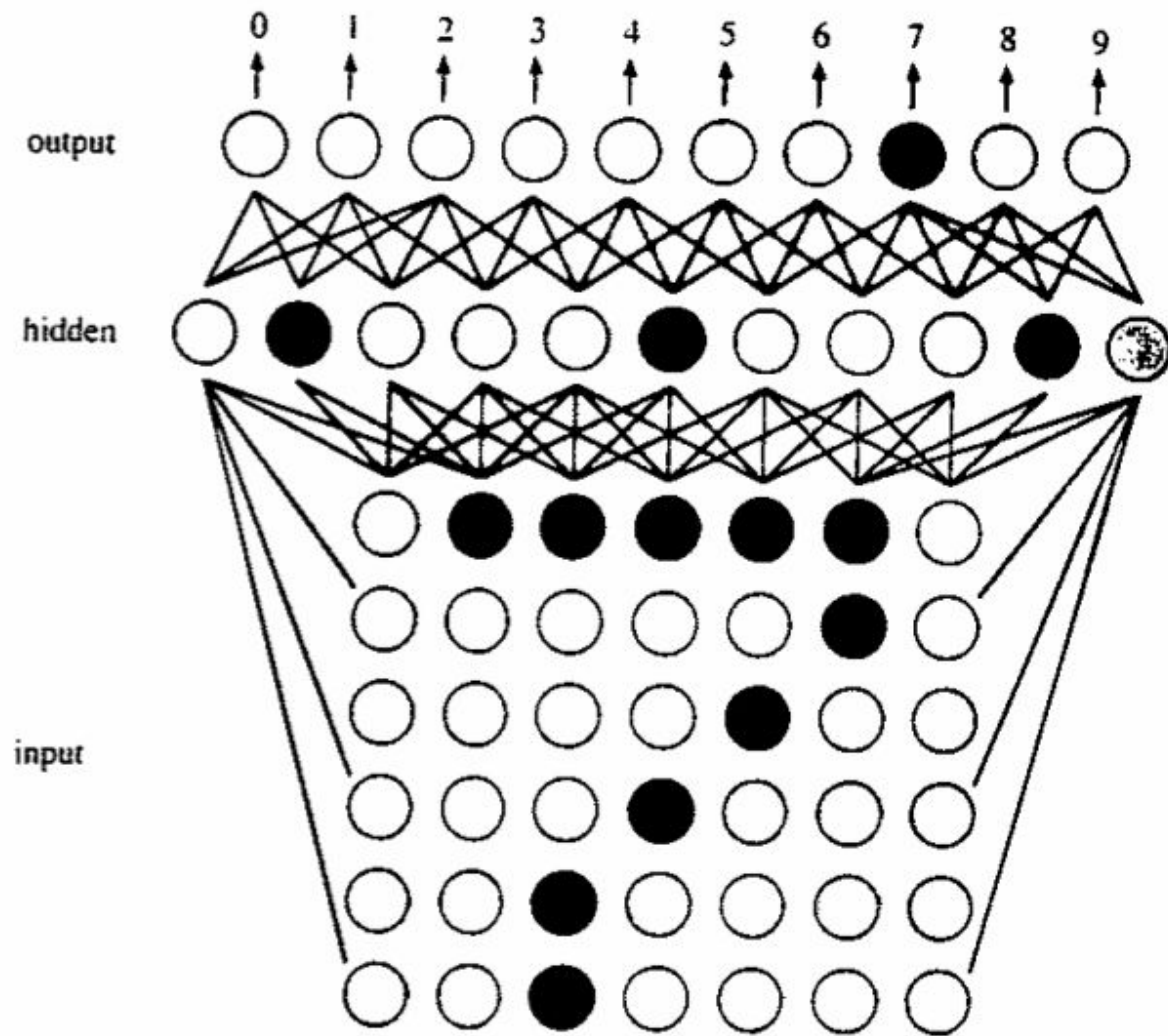
$$\begin{cases} y = 0 & \text{se } u \leq 0 \\ y = 1 & \text{se } u > 0 \end{cases}$$

Reconhecimento de dígitos

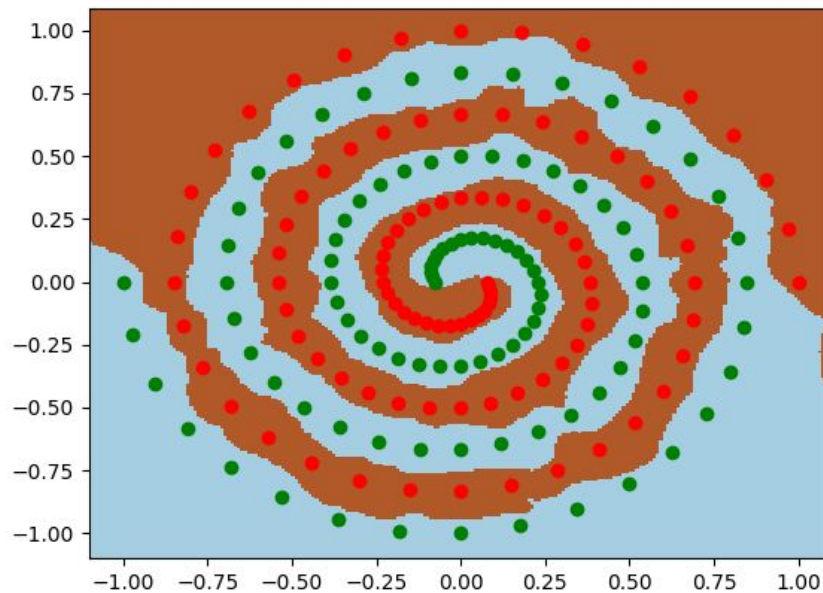
- Camada de entrada?
- Camada de saída?
- Quantas camadas ocultas?
- Quantos neurônios por camada?



Papel das camadas ocultas



Função de Ativação

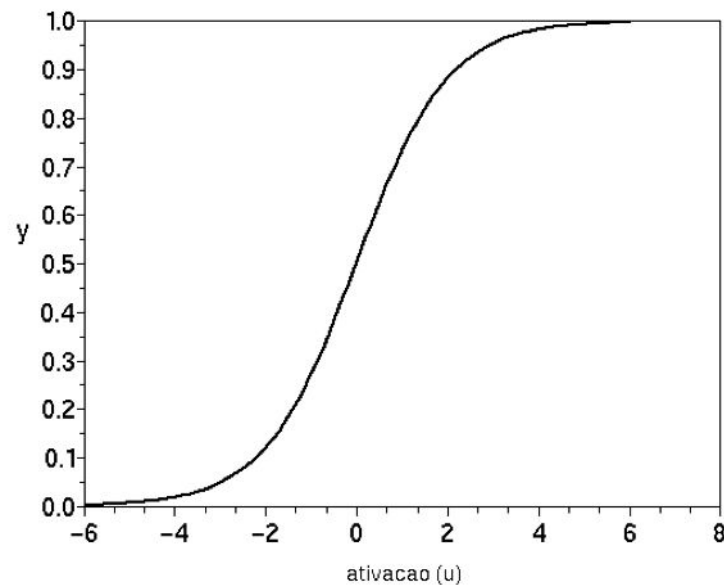


Função de Ativação MLP

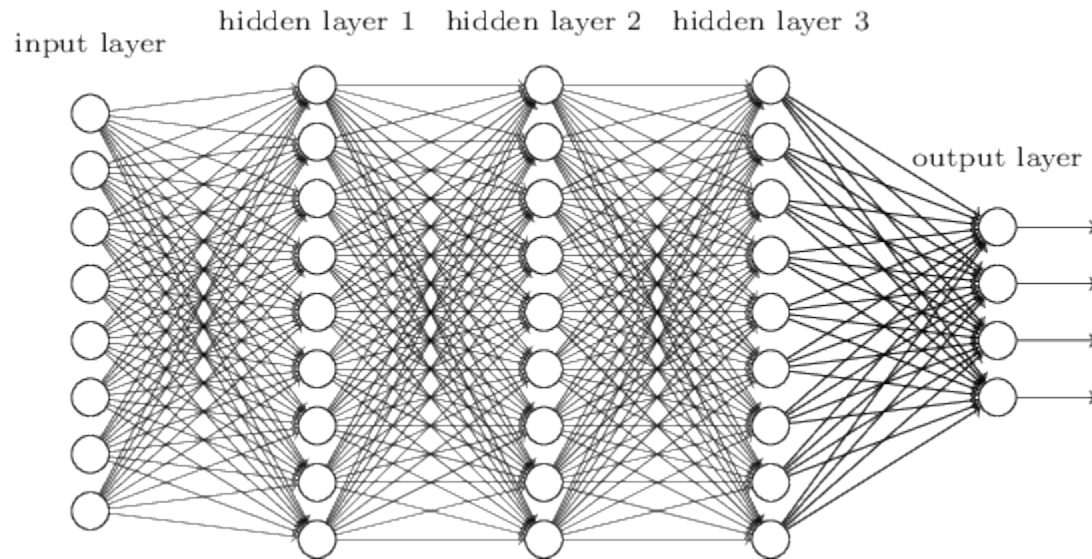
$$y_i(t) = \frac{1}{1 + \exp(-u_i(t))}$$

$$y_i(t) \in (0,1)$$

Função de ativação *Sigmóide Logística*



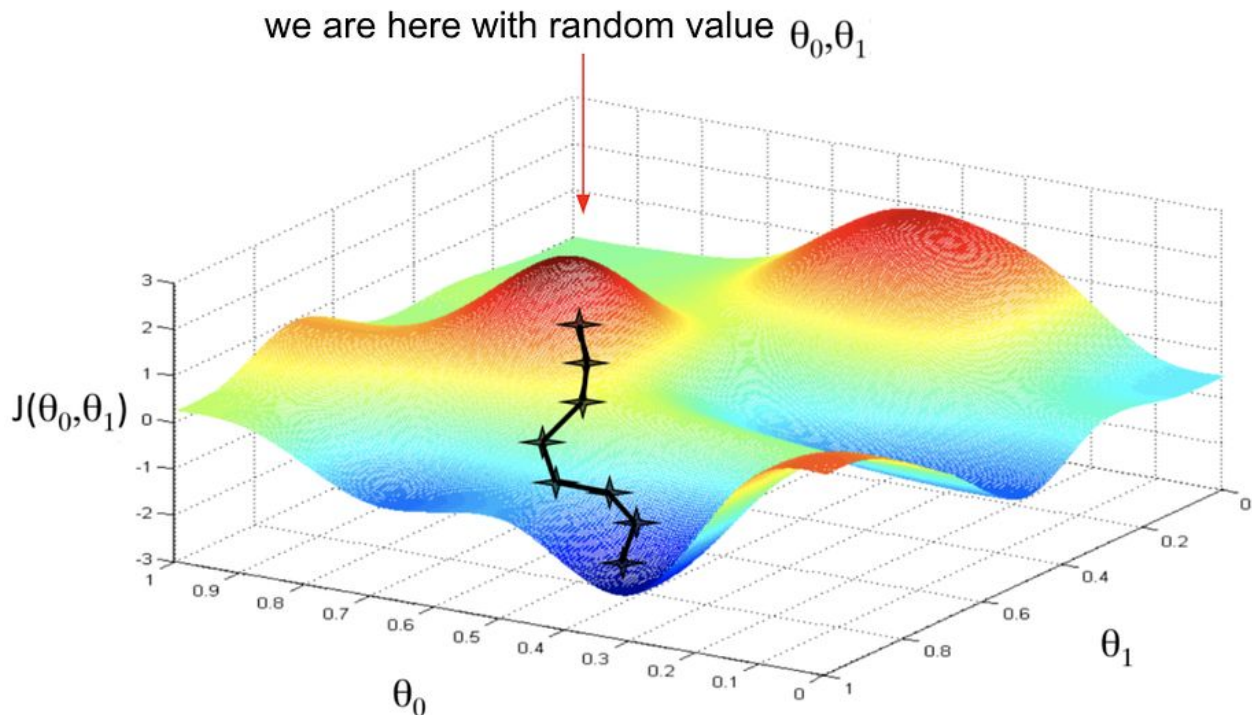
Aprendizagem - e o ajuste?



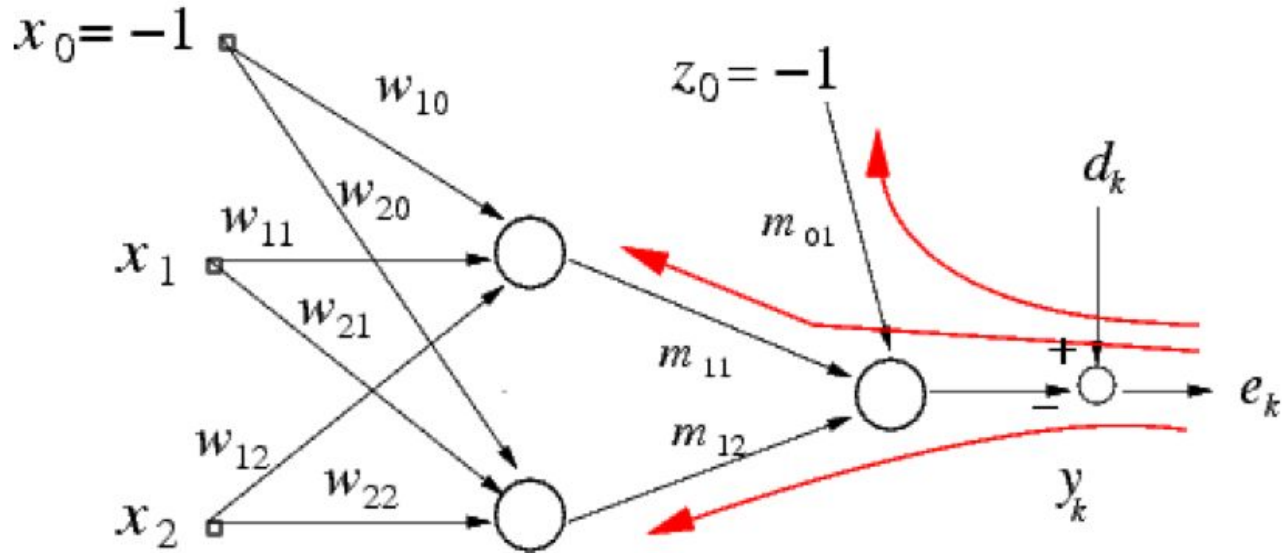
Aprendizagem: problema de otimização

θ = conjunto de pesos

$J(\theta, \theta)$ = cálculo do erro



Backpropagation MLP (Multi-Layered Perceptron)



Função de Ativação MLP

derivada da
sigmoide logística

$$y'_i(t) = \frac{d y_i(t)}{d u_i(t)}$$

$$y'_i(t) = y_i(t)[1 - y_i(t)]$$

